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Lab Number: 1
Class: ELEN 50L
Date: 9/28/2023
Section time: 2:15-5:00

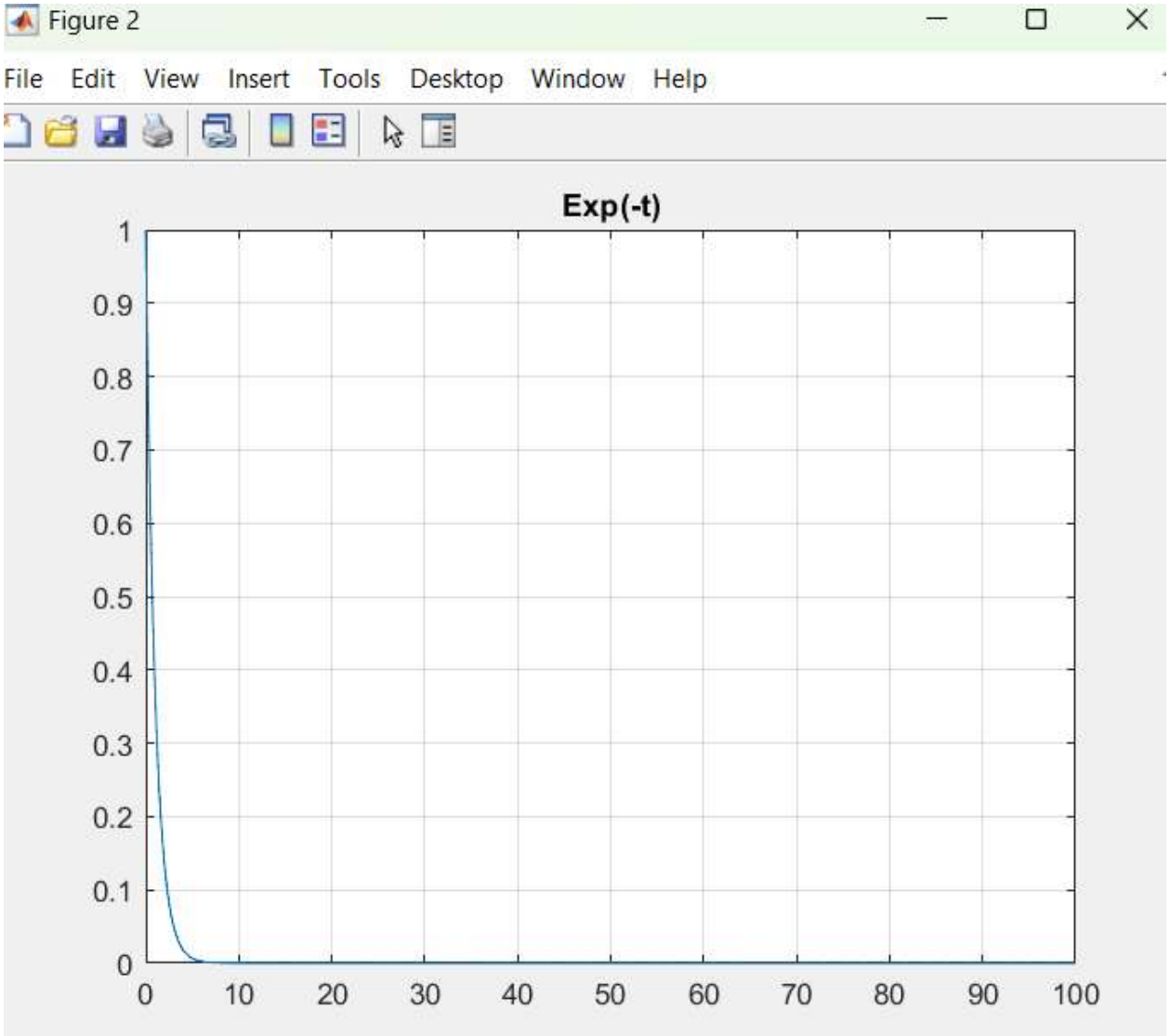
1.

```
(2.3*3+3.0*5+3.7*4)/12

ans =

    3.0583
```

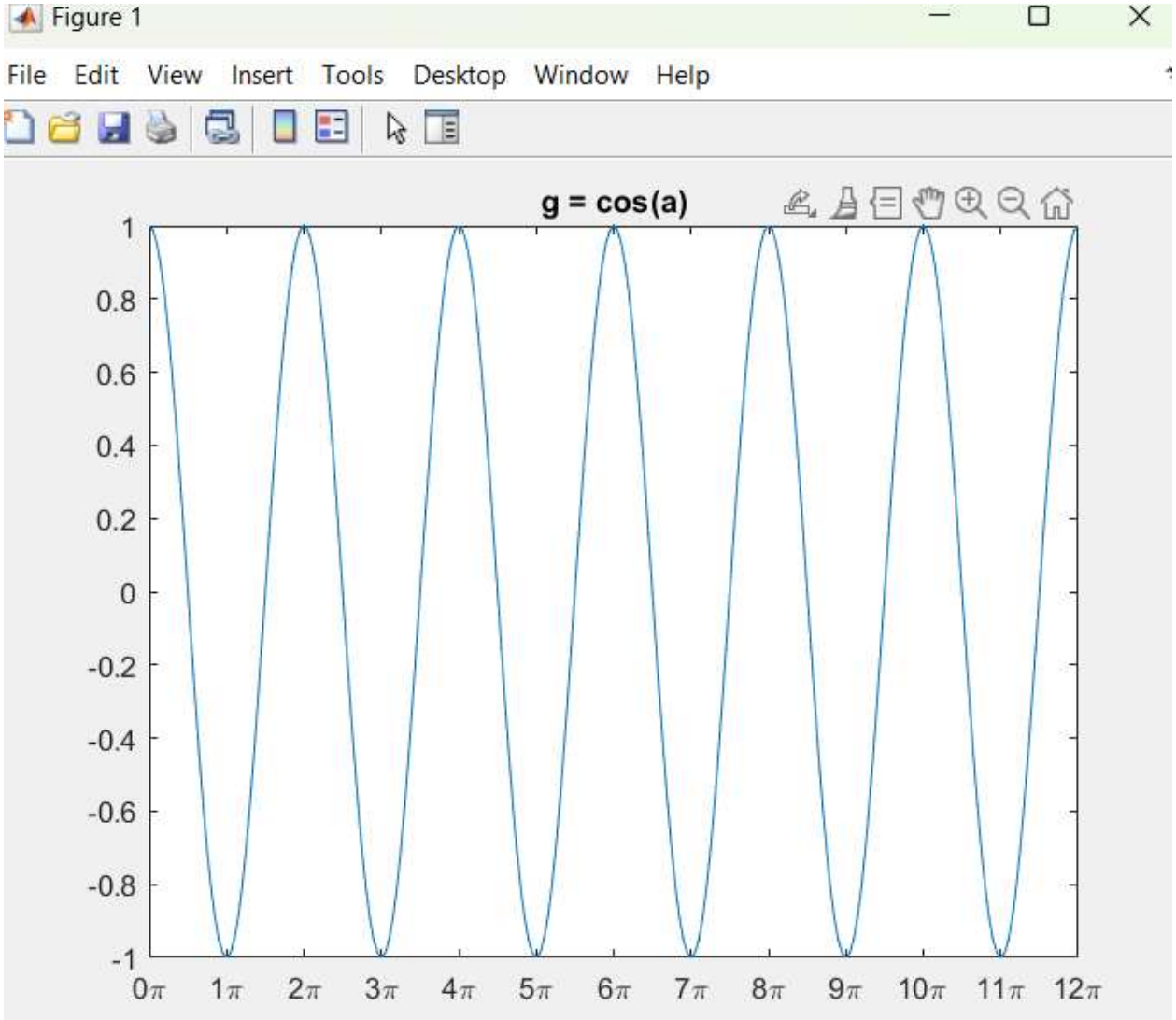
2.

```
cos(2*pi/19*[0:19])

ans =

  Columns 1 through 6

    1.0000    0.9458    0.7891    0.5469    0.2455   -0.0826

  Columns 7 through 12

   -0.4017   -0.6773   -0.8795   -0.9864   -0.9864   -0.8795

  Columns 13 through 18

   -0.6773   -0.4017   -0.0826    0.2455    0.5469    0.7891

  Columns 19 through 20

    0.9458    1.0000
```

3.




```
t = 0:0.01:100;
>> y = exp(-t);
>> plot(t, y);
>> grid on;
```




```
a = pi * (0: .05: 12);
>> y = cos(a);
>> plot(a, y);
>> xlim([0, 12*pi])
>> xticks(pi*[0:1:12]);
>> Labels = string([0:1:12]) + "\pi";
>> xticklabels(Labels);
>>  title('g = cos(a)');
```

4.

```
syms x
f = (exp(-3*x)+10*sin(x+1))/(cos(x+1))
df1 = diff(f,x)
```

```
g = (exp(-2 *x))/(1+exp(-2*x))
df2 = diff(g,x)
h = -4*x^2
If3 = int(h,x,[-2,2])
```

deriv_script

f =

(exp(-3*x) + 10*sin(x + 1))/cos(x + 1)

df1 =

(10*cos(x + 1) - 3*exp(-3*x))/cos(x + 1) + (sin(x + 1)*(exp(-3*x) + 10*sin(x + 1)))/cos(x + 1)^2

g =

exp(-2*x)/(exp(-2*x) + 1)

df2 =

(2*exp(-4*x))/(exp(-2*x) + 1)^2 - (2*exp(-2*x))/(exp(-2*x) + 1)

h =

-4*x^2

If3 =

-64/3

>>

5.

A = [2,1; 3,2]

```
A =

     2     1
     3     2

>> B = [3,1;2,2]

B =

     3     1
     2     2

>> A1 = A * B,

A1 =

     8     4
    13     7

>> A2 = B * A

A2 =

     9     5
    10     6

>> A3 = (A'* B')'

A3 =

     9     5
    10     6

>> A4 = (B' * A')'

A4 =

     8     4
    13     7
```

A1 and A4 are the same and A2 and A3 are the same.

```
A5 = inv (A * B)

A5 =

    1.7500   -1.0000
   -3.2500    2.0000

>>
>> A6 = inv(A) * inv(B)

A6 =

    1.5000   -1.2500
   -2.5000    2.2500

>> A7 = inv (B * A)

A7 =

    1.5000   -1.2500
   -2.5000    2.2500

>> A8 = inv(B) * inv(A)

A8 =

    1.7500   -1.0000
   -3.2500    2.0000


%Inverted


A1 * A5

ans =
```

```
    1.0000   -0.0000
    0.0000    1.0000

>> A2 * A6

ans =

     1     0
     0     1

>> A3 * A7

ans =

    1.0000    0.0000
    0.0000    1.0000

>>  A4 * A8

ans =

    1.0000         0
   -0.0000    1.0000


%End Inverted

>> A1*(A*B)

ans =

   116    60
   195   101

>> (A*B)*A1

ans =

   116    60
   195   101
```

6.

```
C = [1,0,1; 3,3,4; 2,2,3]

C =

     1     0     1
     3     3     4
     2     2     3

>> S = [10;12;5]

S =

    10
    12
     5

>> V = inv(C)*S

V =

   19.0000
   -3.0000
   -9.0000

>> C*V

ans =

   10.0000
   12.0000
    5.0000
```

7.

```
clear all;
close all;
Vs = 12;
Is1 = 4e-3;
```

```
Is2 = 2e-3;
R1 = 1e3;
R2 = 1e3;
R3 = 1e3;
R4 = 1e3;
R5 = 1e3;
R6 = 1e3;
syms Va Vb Vc Vd
eq1 = (Va-Vb)/R1 - Is1 + (Vd-Vc)/R2 + Vd/R5 + Vd/(R3+R6) == 0;
eq2 = (Vb-Va)/R1 + Is2 + (Vc-Vd)/R2 + Vc/R4 == 0;
eq3 = Vc-Vb == Vs;
eq4 = Vd-Va == 2*Vc;
eqns = [eq1, eq2, eq3, eq4];
vars = [Va, Vb, Vc, Vd];
[C, S] = equationsToMatrix(eqns, vars)
V = inv(C) * S
```

```
>> Circuit_script2

C =

[ 1/1000, -1/1000, -1/1000,   1/400]
[-1/1000,  1/1000,   1/500, -1/1000]
[      0,     -1,      1,      0]
[     -1,      0,     -2,      1]


S =

 1/250
-1/500
    12
     0


V =

 -4
-10
  2
  0
```